Math 526, F08
Quiz 8

Name: Answer
SSN: xxx-xx-

Instructions: **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (10 points) True or False.

(a) The determinant of $\mathbf{I} + \mathbf{A}$ is $1 + det\mathbf{A}$.

False

(b) The determinant of $\mathbf{ABC}$ is $|\mathbf{A}||\mathbf{B}||\mathbf{C}|$.

True

(c) The determinant of $4\mathbf{A}$ is $4|\mathbf{A}|$.

False

(d) $det(\frac{1}{2}\mathbf{A}) = -\frac{1}{16}$ if $\mathbf{A}$ is a 4×4 matrix with $det\mathbf{A} = -1$.

True

(e) The determinant of $\mathbf{AB} - \mathbf{BA}$ is zero.

True

Problem 2. (10 points) Find the determinant of $\mathbf{A}$ where $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix}$ using the co-factor expansion.

Solution:

$$|A| = 1 \cdot \begin{vmatrix} 3 & 4 & 5 \\ 4 & 0 & 3 \\ 0 & 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 & 4 \\ 5 & 4 & 0 \\ 2 & 0 & 0 \end{vmatrix}$$

$$= 1 \cdot \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix} - 2 \cdot 2 \cdot \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix}$$

$$= -16 - 4 \cdot (-16)$$

$$= -16 + 64$$

$$= 48.$$